Chapter 9d: Applications of Fixed Point Theorems

Nonlinear Analysis and Fixed Point Theory

Based on H. K. Pathak (2018)

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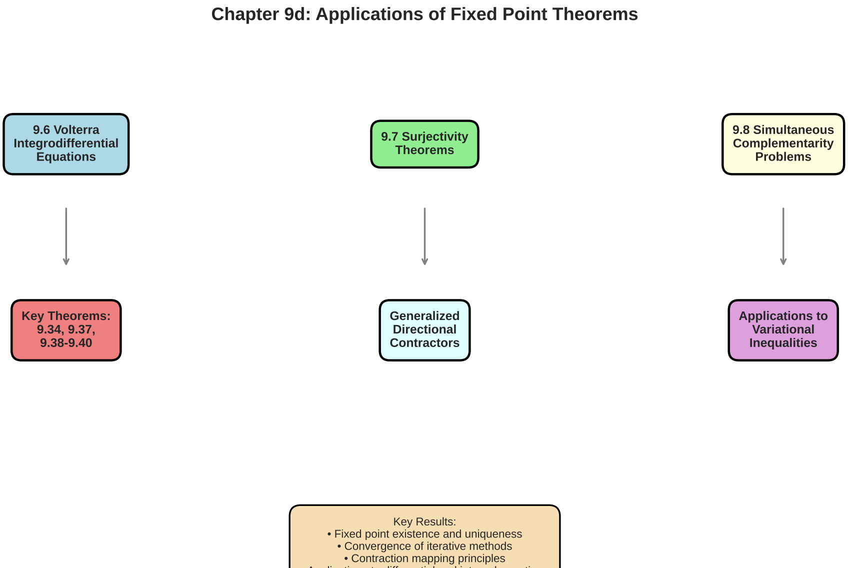


Key Sections in Chapter 9d

1. **Section 9.6:** Applications to Abstract Volterra Integrodifferential Equations
2. **Section 9.7:** Applications to Surjectivity Theorems
3. **Section 9.8:** Applications to Simultaneous Complementarity Problems
4. **Section 9.9:** Generalized Directional Contractors
5. **Exercises:** 9.1 - 9.7

Main Focus: Practical applications of fixed point theorems to important problems in analysis

Section 9.6: Volterra Integrodifferential Equations

Background: Volterra integral equations describe many physical phenomena including

- Hereditary systems
- Systems with memory effects
- Viscoelastic materials
- Population dynamics

Standard Volterra Equation:

$$x(t) = x_0 + \int_0^t a(s)x(s)ds + \int_0^t a(s)\left(\int_0^s b(r)x(r)dr\right) ds + \cdots$$

Key Lemmas for Volterra Equations

Lemma 9.4 (Common Fixed Point)

Let T_1 and T_2 be maps on complete metric space X. If there exists positive integer m and $k < 1$ such that

$$d(T_1^m x, T_2^m y) \leq kd(x, y) \quad \forall x, y \in X$$

then T_1 and T_2 have a unique common fixed point.

Lemma 9.5 (Pachpatte's Inequality)

Let $x(t), a(t), b(t), c(t)$ be real-valued nonnegative continuous functions. If

$$x(t) \leq x_0 + \int_0^t a(s)x(s)ds + \cdots$$

then

$$x(t) \leq x_0 \left(1 + \int_0^t a(s) \exp\left(\int_s^t a(r)dr\right) ds\right)$$

Theorem 9.34: Solution of Volterra Equations

Theorem 9.34 (Pathak)

Suppose conditions A_1–A_3 are satisfied. Then for $u_0 \in B$, the initial value problems

$$\begin{cases} (Fu)(t) = H(T, t_0, f, g, h, K_1, K_2) & t_0 \leq t < \infty \\ (\bar{F}u)(t) = H(T, t_0, \bar{f}, \bar{g}, \bar{h}, \bar{K}_1, \bar{K}_2) & t_0 \leq t < \infty \end{cases}$$

have a unique common mild solution $u(t)$ in $C([t_0, t_1], B)$ for $t \geq t_0$, where t_1 is arbitrary with $t_1 > t_0$.

Proof Idea

- Define norm on Banach space
- Show F and $\bar{F}$ are contraction mappings
- Apply common fixed point theorem

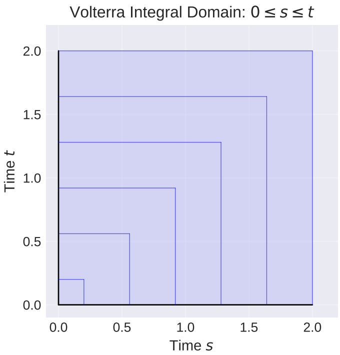
Numerical Example: Volterra Equation

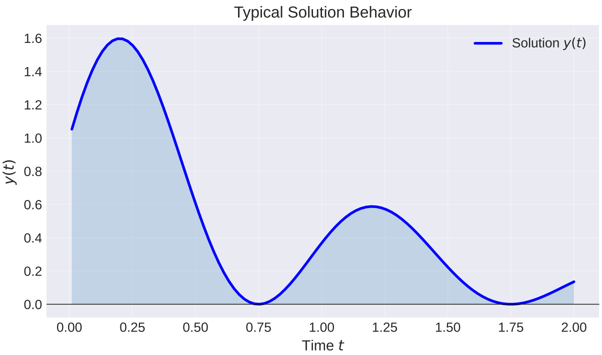
Problem: Solve $x(t) = 1 + \int_0^t x(s)ds$ on $[0, 1]$

```
import numpy as np
from scipy.integrate import odeint

def volterra_example():
    # Kernel: K(t,s) = 1 (exponential kernel)
    def kernel(t, s):
        return 1.0

    # Iterative solution
    x = lambda t: 1.0  # Initial guess

    for iteration in range(10):
        # Update using Picard iteration
        x_new = lambda t: 1.0 + np.trapz(
            [x(s) for s in np.linspace(0, t, 50)],
            np.linspace(0, t, 50))
        x = x_new

    # Exact solution: x(t) = e^t
    t_vals = np.linspace(0, 1, 100)
    x_exact = np.exp(t_vals)

    return t_vals, x_exact
```

Listing 1: Volterra Equation Solver

Volterra Equations: Graphical Illustration





Left: Integration domain for Volterra equations
Right: Typical solution behavior with exponential decay

Section 9.7: Applications to Surjectivity Theorems

Problem: Under what conditions is operator $P : \mathcal{D}(P) \to Y$ surjective?

Definition

A mapping $P : X \to Y$ is **surjective** if for every $y \in Y$, there exists $x \in X$ such that $P(x) = y$.

- Classical surjectivity theorems apply only to linear operators
- Extension to nonlinear operators requires new techniques
- Fixed point methods provide powerful tools

Directional Contractors

Definition

Let $P : \mathcal{D}(P) \subset X \to Y$. The mapping $\Gamma : \mathcal{D}(P) \to \mathcal{D}(P)$ is a **directional contractor** for P if there exists $q = q(P) < 1$ and $\varepsilon = \varepsilon(x, y) \in (0, 1]$ such that

$$\|P(x + \varepsilon\Gamma(x)y) - Px - \varepsilon y\| \leq q\varepsilon\|y\|$$

Definition

Γ is a **generalized directional contractor** if

$$\|P(x + \varepsilon\Gamma(x)y) - Px - \varepsilon y\| \leq q\varepsilon c(\|x\|)\|y\|$$

where $c : [0, \infty) \to (0, q^{-1/2})$ is nonincreasing.

Theorem 9.37: Surjectivity via Directional Contractors

Theorem 9.37

Let X and Y be two Banach spaces. A nonlinear map $P : \mathcal{D}(P) \subset X \to Y$ which has closed graph and a bounded generalized directional contractor Γ is surjective.

Proof Strategy

1. Define metric $\rho(x, y) = \max\{\|x - y\|, (1 + q^{1/2})^{-1}\|Px - Py\|\}$ on $\mathcal{D}(P)$
2. Show $(\mathcal{D}(P), \rho)$ is complete metric space
3. Construct contractive iterative process
4. Use fixed point theorem to prove existence

Application to Specific Problems

Example 1: Nonlinear Integral Equation

$$u(t) = h(t) + \int_0^t K(t,s)u(s)ds$$

Surjectivity: For given $h \in L^p[0, T]$, does a solution exist?

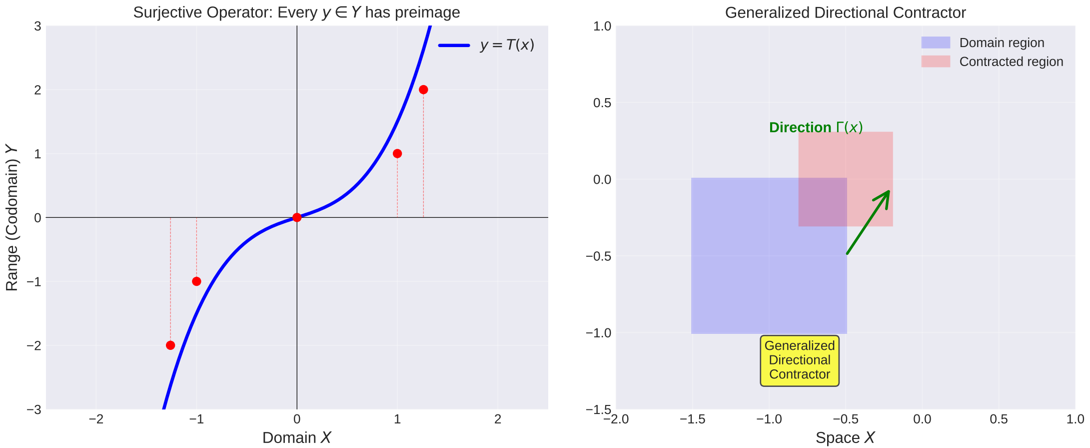
Example 2: Boundary Value Problem

$$-u'' = f(t, u(t)), \quad u(0) = u(1) = 0$$

Question: Is the Green's operator surjective onto appropriate function spaces?

- Use directional contractor to establish surjectivity
- Guarantee existence of solutions for all right-hand sides
- Extend to nonlinear settings

Surjectivity and Fixed Points



Left: Surjective operator mapping domain to full codomain
Right: Generalized directional contractor with contraction property

Section 9.8: Complementarity Problems

Background: Complementarity problems arise in

- Optimization and mathematical programming
- Mechanics (contact problems, friction)
- Economics (equilibrium problems)
- Game theory
- Traffic flow and network analysis

Historical Note: First studied systematically in 1960s by Lemke and Howson

Explicit Complementarity Problem (ECP)

Definition

Explicit Complementarity Problem (ECP): Find $x_0 \in K$ such that $f(x_0) \in K^*$ and $\langle x_0, f(x_0) \rangle = 0$, where

- (E, E^*) is a dual system of locally convex spaces
- K is a closed convex cone in E
- $K^* = \{u \in E^* : \langle x, u \rangle \geq 0 \text{ for all } x \in K\}$ is dual cone
- $f : K \to E^*$ is mapping

Notation: $0 \leq x_0 \perp f(x_0) \geq 0$ (orthogonal complementarity)

Implicit Complementarity Problem (ICP)

Definition

Implicit Complementarity Problem (ICP): Find $x_0 \in K$ such that $\langle x_0, g(x_0) \rangle = 0$ where

- $g : K \to E^*$ is a mapping
- Solution must satisfy additional constraints

Relationship Between ECP and ICP

- ICP is more general than ECP
- ICP can model implicit functional constraints
- Both require finding orthogonal pairs in cone-dual cone structure

Linear Complementarity Problem (LCP)

Definition

Linear Complementarity Problem (LCP): Find $x \in \mathbb{R}^n$ such that

$$\begin{cases} 0 \le x \\ 0 \le Mx + q \\ \langle x, Mx + q \rangle = 0 \end{cases}$$

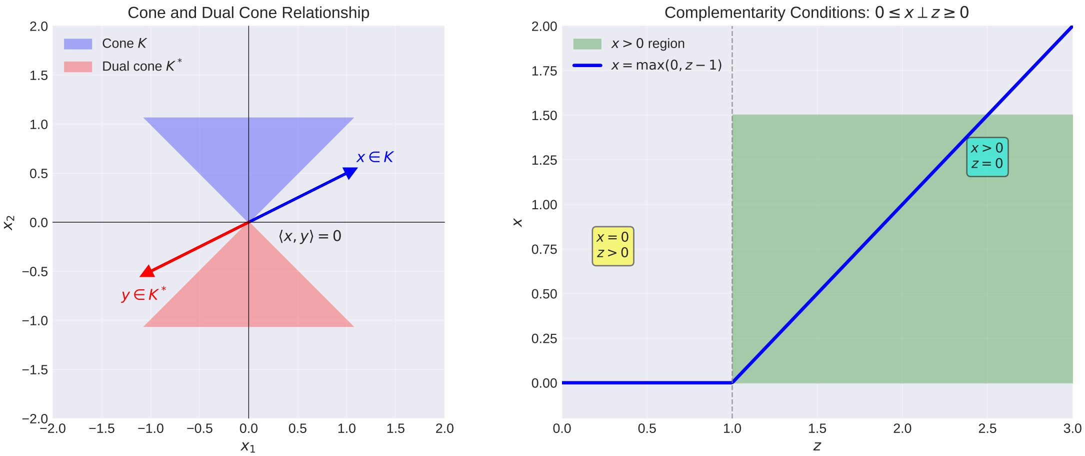
where $M \in \mathbb{R}^{n \times n}$ and $q \in \mathbb{R}^n$.

Notation: $\mathrm{LCP}(M, q)$

Special Cases

- If M is positive definite: unique solution
- If M is monotone: solution may not be unique but existence guaranteed
- General M: solution may not exist

Complementarity Problem Geometry



Left: Cone and dual cone relationship with orthogonality

Right: Complementarity condition: $x \geq 0 \perp z \geq 0$

Numerical Solution: LCP Example

Problem: Solve LCP(M, q) where

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad q = \begin{pmatrix} -3 \\ -3 \end{pmatrix}$$

```
import numpy as np

def lemke_method(M, q, max_iter=1000):
    n = len(q)
    x = np.zeros(n)
    s = -q.copy()

    for iteration in range(max_iter):
        # Find entering variable
        if np.all(s >= 0):
            return x  # Feasible solution found

        j = np.argmin(s)  # Most negative slack

        # Update using complementarity
        x[j] += s[j] / M[j, j]
        s = -q - M @ x

    return x

# Solve
x_sol = lemke_method(M, q)
```

Definition 9.9: Generalized Directional Contractor

Definition

Let X and Y be Banach spaces, $P : \mathcal{D}(P) \subset X \to Y$ a nonlinear operator from linear subspace $\mathcal{D}(P)$ of X to Y, and $\Gamma : \mathcal{D}(P) \to \mathcal{D}(P)$ a bounded linear operator. A positive number $q = q(P) < 1$ exists such that for any $x \in \mathcal{D}(P)$ and $y \in Y$, there exist $\varepsilon = \varepsilon(x, y) \in (0, 1]$ and a nonincreasing function $c : [0, \infty) \to (0, q^{-1/2})$ satisfying

$$\|P(x + \varepsilon\Gamma(x)y) - Px - \varepsilon y\| \leq q\varepsilon c(\|x\|)\|y\|$$

Then $\Gamma(x)$ is called a **generalized directional contractor** for P at $x \in \mathcal{D}(P)$.

Properties of Directional Contractors

Theorem

For a generalized directional contractor Γ*:*

1. *If* $\Gamma(x)y = 0$, *then* $y = 0$ *(injectivity)*
2. Γ *is bounded:* $\|\Gamma(x)\| \leq B$ *for some constant* $B > 0$
3. $\Gamma(x)$ *measures the "direction" of steepest descent for* P
4. *Inverse operator approximated by directional contractor*

Key Observation

Every inverse Fréchet derivative is a directional contractor, and every generalized directional contractor is a directional contractor.

Applications of Directional Contractors

Application 1: Surjectivity

Use directional contractor to prove surjectivity of operators with closed graph

Application 2: Implicit Function Theorem

Establish implicit function theorems for operators admitting directional contractors

Application 3: Variational Inequalities

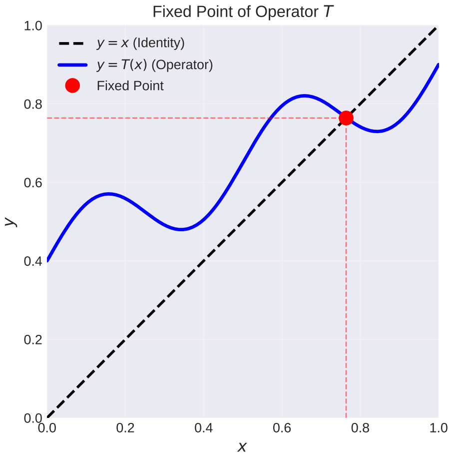
Solve variational inequalities of form:

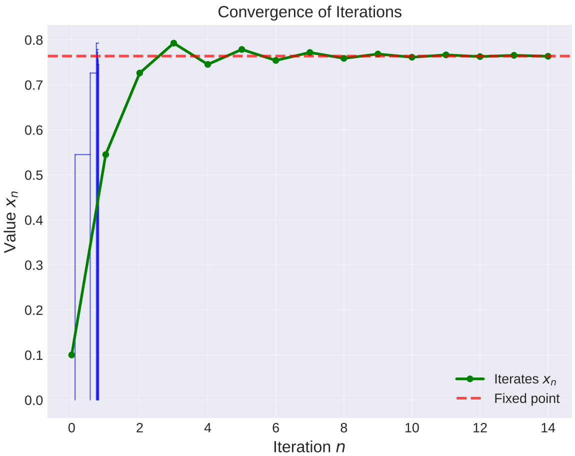
$$\langle P(x), y - x \rangle \geq f(y) - f(x) \quad \forall y \in K$$

Application 4: Nonlinear Equations

Solve $P(x) = 0$ where P has directional contractor

Fixed Point Geometry





Left: Intersection of function with identity line defines fixed point
Right: Iterative convergence to fixed point under contraction

Banach Contraction Mapping Principle

Theorem (Banach)

Let (X, d) be a complete metric space and $T : X \to X$ a mapping such that

$$d(T(x), T(y)) \leq kd(x, y) \quad \forall x, y \in X$$

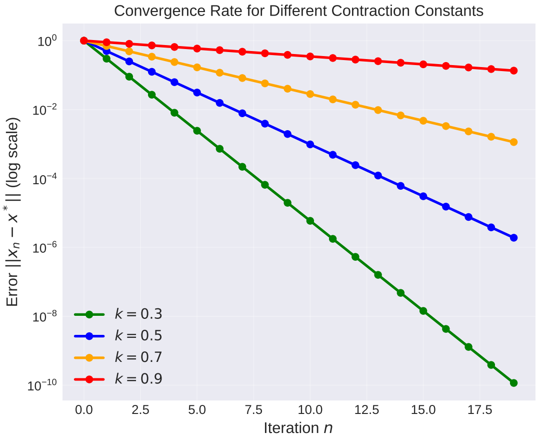
for some $k \in [0, 1)$. Then:

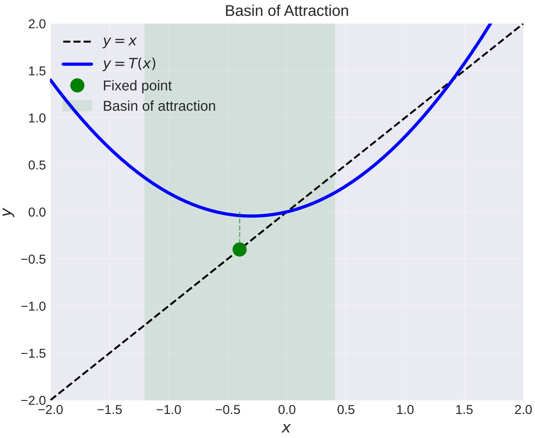
1. *T has a unique fixed point $x^* \in X$*
2. *The sequence $\{x_n\}$ defined by $x_{n+1} = T(x_n)$ converges to x^* for any $x_0 \in X$*
3. *Error estimate: $d(x_n, x^*) \leq k^n d(x_0, x^*)$*

Why Important for Chapter 9d?

All applications rely on this fundamental principle by constructing appropriate operators and metrics

Contraction Mapping Convergence Rates





Left: Error decreases exponentially with iteration number
Right: Basin of attraction around fixed point

Conditions for Fixed Point Existence I

Condition 1: Complete Metric Space

(X, d) must be complete. Alternatively, use Banach spaces with

- Closed subsets (complete)
- Natural metrics

Condition 2: Contraction Property

Need $d(T(x), T(y)) \leq kd(x, y)$ with $k < 1$. This can be:

- Uniform contraction
- Lipschitz continuous mapping
- Operator norm condition

Conditions for Fixed Point Existence II

Condition 3: Appropriate Metric

Choice of metric is crucial:

- Standard metric: $d(x, y) = \|x - y\|$
- Weighted metric: $\rho(x, y) = w(x, y)\|x - y\|$
- Custom metric tailored to problem

Conditions for Fixed Point Existence III

Metric Construction in Chapter 9d

For surjectivity problems:

$$\rho(x, y) = \max\{\|x - y\|, (1 + q^{1/2})^{-1}\|Px - Py\|\}$$

This metric makes $(\mathcal{D}(P), \rho)$ complete when original space isn't!

Fixed Point Methods vs. Optimization Methods

Property	Fixed Point	Newton	Gradient Descent
Convergence Rate	Linear	Quadratic	Sublinear
Derivative Needed	Yes/No	Yes	Yes
Stability	Global	Local	Slow
Complexity/Iter	Low	High	Low
Nonlinear Problems	Yes	Yes	Limited

Fixed Point Advantages

- Works without derivative information
- Global convergence under weak conditions
- Applicable to wide class of problems

Extensions Beyond Chapter 9d

- **Partial Metric Spaces:** Generalization for distance to self not necessarily zero
- **Fuzzy Metric Spaces:** Probabilistic distances between points
- **Multivalued Maps:** Fixed point results for set-valued operators
- **Fractional Operators:** Application to fractional differential equations
- **Variational Inclusions:** Generalizations to inclusion problems
- **Metric-like Spaces:** Weaker distance axioms but stronger theorems

Real-World Applications I

1. Mechanical Systems

- Vibration problems with friction and damping
- Contact mechanics and Coulomb friction
- Elastic-plastic deformations

2. Economic Equilibrium

- Supply-demand equilibrium
- Nash equilibrium in games
- Market clearing conditions

Real-World Applications II

3. Systems with Memory

- Viscoelastic materials
- Biological systems with hereditary effects
- Chemical reactions with memory terms

4. Traffic Flow and Networks

- Network flow problems
- User equilibrium on transportation networks
- Network design optimization

Practical Implementation Example

```python
import numpy as np

def fixed_point_solver(T, x0, tol=1e-8, max_iter=1000):
    """
    Solve x = T(x) using fixed point iteration

    Args:
        T: Function defining the fixed point operator
        x0: Initial guess
        tol: Convergence tolerance
        max_iter: Maximum iterations

    Returns:
        x_star: Fixed point solution
    """
    x = x0
    history = [x.copy()]

    for iteration in range(max_iter):
        x_new = T(x)
        error = np.linalg.norm(x_new - x)
        history.append(x_new.copy())

        if error < tol:
            print(f"Converged in {iteration} iterations")
            return x_new, np.array(history)
```

Theorem 9.34

Volterra Equations: Existence and uniqueness of common mild solution for two operators satisfying contraction property on appropriate Banach space.

Theorem 9.37

Surjectivity: Nonlinear operator with closed graph and bounded generalized directional contractor is surjective onto Y.

Theorems 9.38-9.40

Complementarity: Existence results for explicit and implicit complementarity problems using fixed point theory and monotone operator theory.

Definition 9.9

Generalized Directional Contractor: Extension of classical directional contractor with function-valued contraction constant.

Proof Techniques Used

1. **Metric Construction:** Create custom metrics to ensure completeness and contractivity
2. **Banach Theorem:** Apply fundamental fixed point theorem to constructed operators
3. **A Priori Estimates:** Use inequalities (Pachpatte, Grönwall) to control solutions
4. **Operator Decomposition:** Split problem into parts, apply theory to each
5. **Monotonicity/Accretivity:** Exploit structural properties of operators
6. **Continuous Dependence:** Show stability of solutions with respect to parameters

Exercise 9.1

For the integrodifferential equation

$$\frac{dx}{dt} + \int_0^t x(s)ds = e^{-t}, \quad x(0) = x_0$$

prove existence and uniqueness of solution in $C([0, T])$ using fixed point theorems.

Exercise 9.2

Show that the initial value problem

$$\begin{cases} \frac{du}{dt} = u^{1/3}, \quad t > 0 \\ u(0) = 0 \end{cases}$$

has more than one solution, and discuss which solutions are obtained via fixed point methods.

Exercise 9.3

Consider the Volterra integral equation

$$x(t) = |x(t)|^{1/2}, \quad x(0) = 0$$

Does this have solution? Find solutions explicitly.

Exercise 9.4

Transform the second-order differential equation

$$\frac{d^2x}{dt^2} = f(t, x), \quad x(0) = x_0, \, x'(t_0) = x_1$$

into a Volterra integral equation form suitable for fixed point analysis.

Selected Exercises from Chapter 9d III

Exercise 9.5

For the Fredholm integral equation

$$f(t) = g(t) + \lambda \int_0^1 K(t,s) f(s) ds$$

with $g \in L_2[0,1]$ and $K \in L_2([0,1] \times [0,1])$, prove unique solvability for appropriate λ.

Exercise 9.6

For the nonlinear integral equation

$$v(t) = x(t) - \lambda \int_a^h K(t,s,x(s)) ds$$

establish when it has solution and examine Lipschitz continuity with respect to initial conditions.

Exercise 9.7

Consider nonlinear integral equation

$$v(t) = x(t) - \lambda \int_a^h K(t, s, x(s))ds$$

and show it satisfies fixed point property under appropriate conditions.

Key Takeaways

1. **Fixed Point Theory is Powerful:** Can solve many nonlinear problems without calculus
2. **Metric Engineering:** Choosing right metric is crucial for applying fixed point theorems
3. **Wide Applicability:** Covers integrodifferential equations, surjectivity, complementarity
4. **Theoretical + Practical:** Provides both existence results and computational algorithms
5. **Extensible Framework:** Generalizations to directional contractors, multivalued maps, etc.

Further Reading and Resources

- **Primary Reference:** Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Springer.
- **Complementarity Theory:** Facchinei, F., & Pang, J.-S. (2003). *Finite-Dimensional Variational Inequalities and Complementarity Problems*.
- **Integral Equations:** Gripenberg, G., Londen, S.-O., & Staffans, O. (1990). *Volterra Integral and Functional Equations*.
- **Fixed Point Theory:** Krasnoselskii, M. A. (1964). *Topological Methods in the Theory of Nonlinear Integral Equations*.

Online Resources:

- MathOverflow and Mathematics Stack Exchange for discussions
- Springer Link and Google Scholar for recent papers

Applications of Fixed Point Theorems

Successfully covered:

- Volterra integrodifferential equations
- Surjectivity theorems with directional contractors
- Complementarity problems
- Generalized directional contractors
- Practical applications and algorithms

Ready for Chapter 10: Multifunction Integral Inclusions